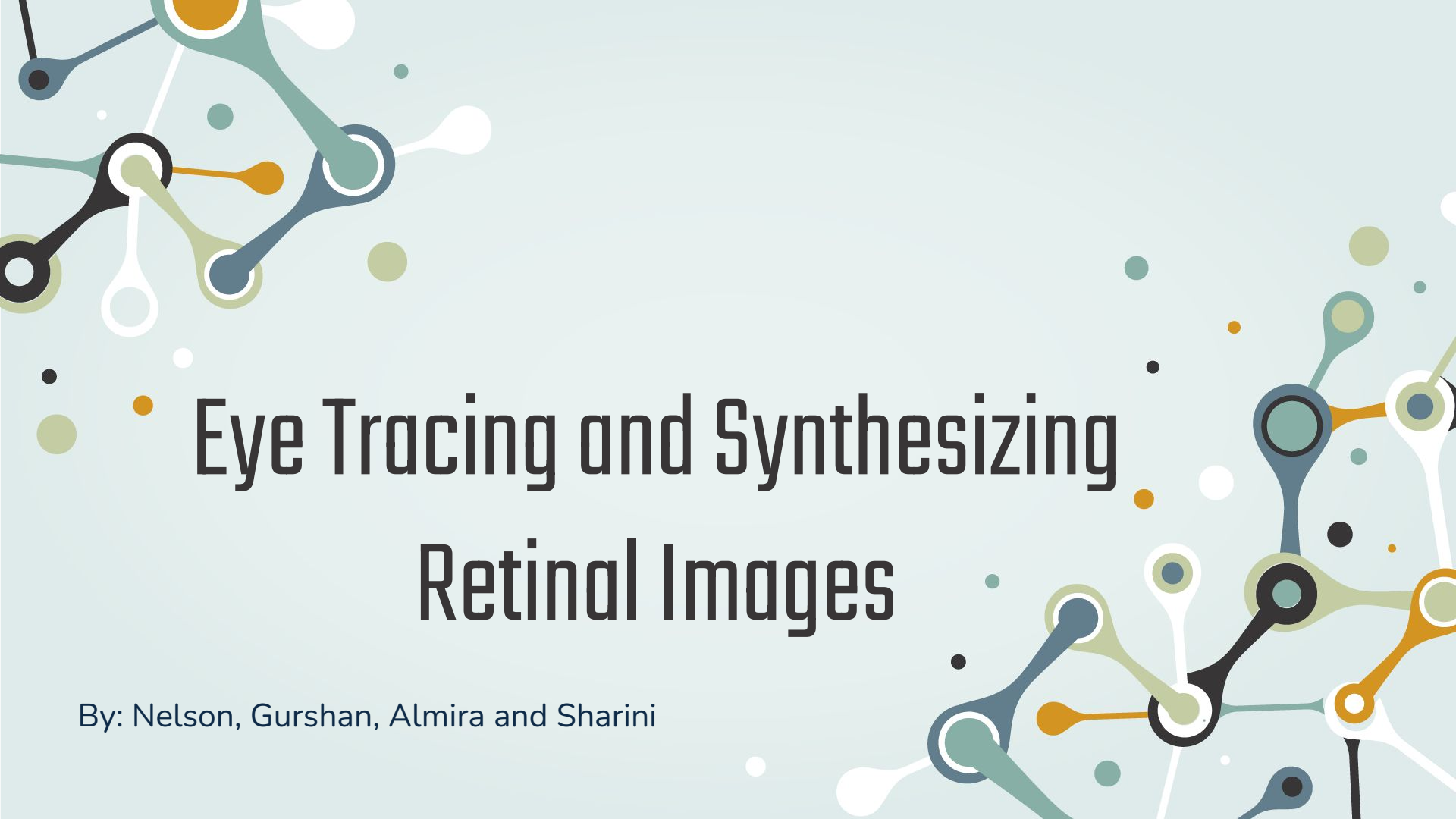


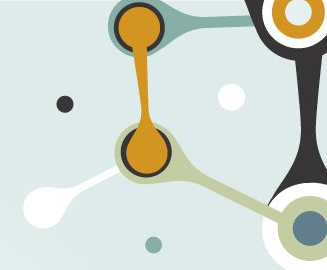


Eye Tracing and Synthesizing Retinal Images

By: Nelson, Gurshan, Almira and Sharini



Problem & Goal



The problem:

Predictive models that determine qualities of different diseases are limited by a small amount of patient data.

Our solution:

Generate more data based on a disease prompt that can be used to train **larger disease classification models**.

Approaches and Methodologies

01

Data analysis & Interpretation

The initial step is to create a well-functioning system that contains and organises the data effectively.

02

Generative Model

The identified data patterns are used to train the model in order to generate new, synthetic data within the possible range. Ultimately, this is where we 'create' new patients.

03

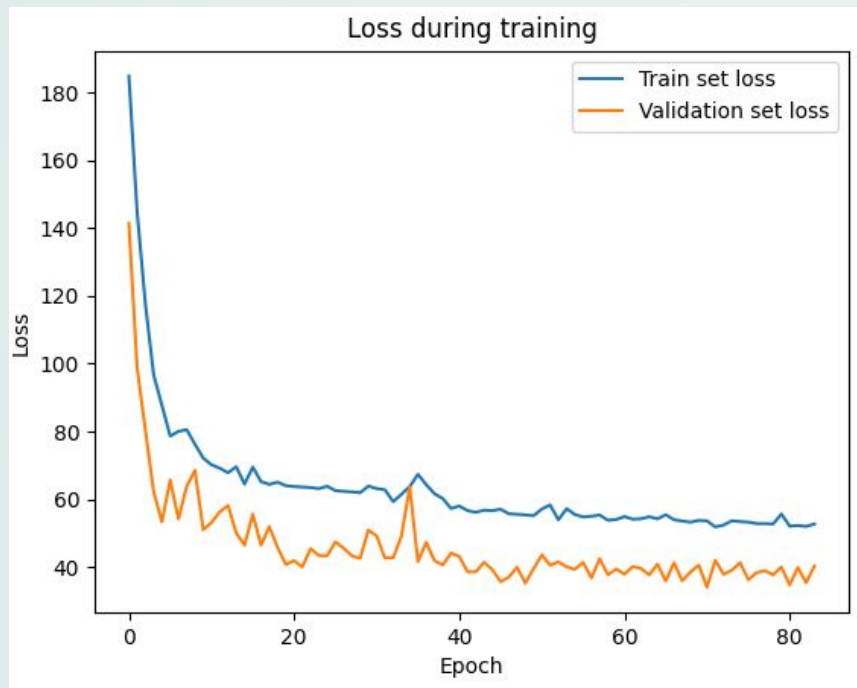
Downstream -- Decision Tree

We then use those new 'patients' to improve current models. To represent an example of this, we trained a decision tree classifier to output a disease based on an input of eye trace statistics.

Data processing

- Systemised everything inside one data set
 - parsed filenames and extracted metadata
 - combined and reordered the columns and rows in order to display data systematically in the dataframe
- Implemented custom PyTorch Dataset class to help us leverage industry-standard libraries
 - We can leverage multi-GPU and multi-node training libraries and modern PyTorch Transformer tools
- Selecting and Dropping the incomplete data
- Selecting data
 - Focusing on numeric data format for the current stage of the project

The Generator



We fit a standard Encoder-Decoder transformer to the eye traces taken from patients. We find that the model seems to generalize well to other MCI patients!

Necessary Stats:

- 70% Train split
- 20% Test split

```
get_avg_loss(model, test_dl).item()
```

```
tensor(65.0591, device='cuda:0')
```

Loss
Looks
similar!



The Decision Tree



Trained a decision tree classifier that takes statistics of eye trace videos and classifies them into disease categories.

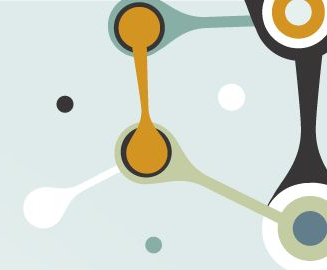
- ~ 70% accuracy
- 16 summary statistics of an eye trace video used to train
- 11 levels

Training/test split:

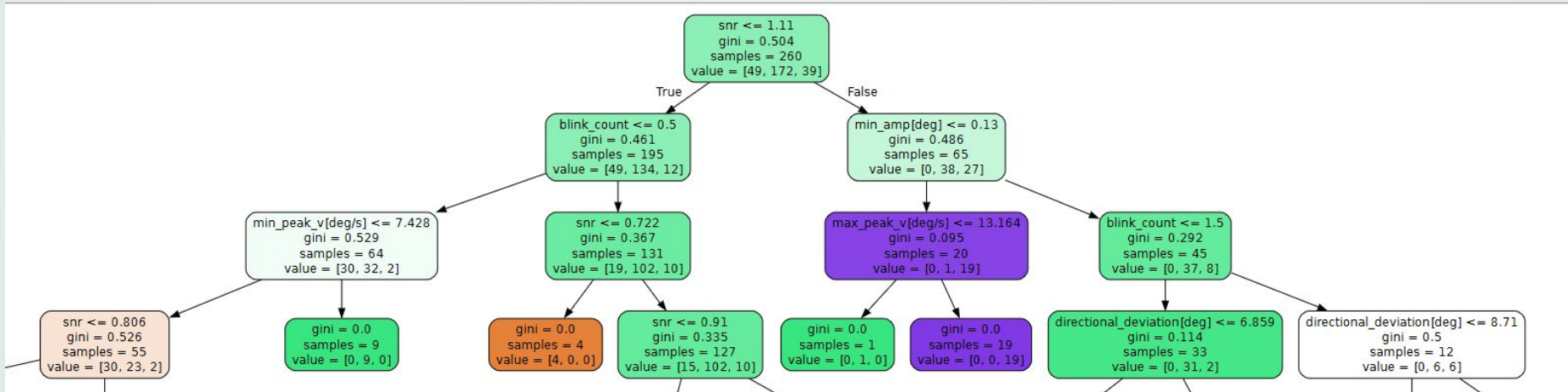
- $\frac{1}{3}$ test, $\frac{2}{3}$ training



Connecting Decision Trees to disease classification



- Peak velocity correlated with disease
- Measuring mTBI with vertical and horizontal saccades



Eye movement can be used as an important indicator of a variety of medical diseases from the correlation of eye movement. For example for mTBI recordings of eye movements can develop into eye tracing images focusing on measuring vertical and horizontal saccades in order to quantify the severity of TBI which can be represented by the classification categories above such as blink count affecting saccade or its minimum amplitude

Background on diseases we have information on

mTBI (Mild traumatic Brain Injury): Traumatic brain injury is usually caused by a blow or other traumatic injury to the head or body.

→Saccadic reaction time can be used to monitor patients with mTBI and to observe abnormalities in saccadic reaction time, reaction times in mTBI patients are usually prolonged/delayed post injury.

MS (Multiple Sclerosis): Multiple sclerosis is a neurodegenerative disease where the myelin sheaths of a nerve are damaged in the brain and spinal cord.

→MS can cause eye movement abnormalities as they can develop a type of saccadic disorder called INO or internuclear ophthalmoparesis which causes the slowing of adducting eye during horizontal saccades.

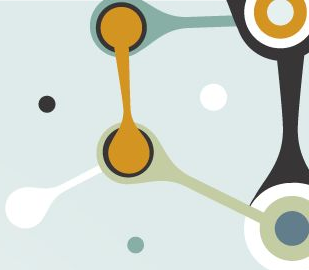
MCI (Mild Cognitive Impairment): Mild cognitive impairment is a condition where people have memory/ thinking problems and eye tracking technology can be used to measure cognitive impairment.

→ Patients with MCI complete the saccadic eye movement paradigms and if patients generate a high proportion of erroneous saccades, it tends to indicate MCI.

Saccades : Rapid eye movement between two fixed points and can be measured from eye traces which can help predict diseases. Eye movement function can be detected by saccadic movement using the eye traces



Further Prospects



- Adding the second step for track B - A generative model that creates eye trace videos
- Implement a **Dataset** abstraction around the generator
 - To help improve current AI models with more data
- Currently, our generative model requires a first datapoint
 - We currently randomly sample from existing data
 - Future work on diffusion models to generate plausible starting points

The background is a light blue-grey color. It features several abstract geometric elements: thick lines in orange, teal, and white that connect circular nodes. Some nodes are solid circles, while others are white circles with black outlines. Scattered throughout the background are numerous small circles in white, black, teal, and orange. The overall style is modern and minimalist.

Thank you!



Questions?